

Spring 1995

Problem 1 (Sp95). Prove that the sequence of functions $f_n(x) = \cos nx$ has no uniformly convergent subsequence.

Problem 2 (Sp95). Let A be the 3×3 matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

Determine all real numbers a for which the limit $\lim_{n \rightarrow \infty} a^n A^n$ exists and is nonzero (as a matrix).

Problem 3 (Sp95). Let n be a positive integer and $0 < \theta < \pi$. Prove that

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{z^n}{1 - 2z \cos \theta + z^2} dz = \frac{\sin n\theta}{\sin \theta}$$

where the circle $|z| = 2$ is oriented counterclockwise.

Problem 4 (Sp95, Sp19). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded continuously differentiable function. Show that every solution of $y'(x) = f(y(x))$ is monotone.

Problem 5 (Sp95). Suppose that R is a subring of a commutative ring S and that R is of finite index n in S . Let m be an integer that is relatively prime to n . Prove that the natural map $R/mR \rightarrow S/mS$ is a ring isomorphism.

Problem 6 (Sp95). Let $f, g : [0, 1] \rightarrow [0, \infty)$ be continuous functions satisfying

$$\sup_{0 \leq x \leq 1} f(x) = \sup_{0 \leq x \leq 1} g(x)$$

Prove that there exists $t \in [0, 1]$ with $f(t)^2 + 3f(t) = g(t)^2 + 3g(t)$.

Problem 7 (Sp95). Suppose that $W \subset V$ are finite dimensional vector spaces over a field, and let $L : V \rightarrow V$ be a linear transformation with $L(V) \subset W$. Denote the restriction of L to W by L_W . Prove that $\det(I - tL) = \det(I - tL_W)$.

Problem 8 (Sp95). Let $p(x)$ be a polynomial with real coefficients and with leading coefficient 1. Suppose that $p(0) = -1$ and that $p(x)$ has no complex zeros inside the unit circle. Prove that $p(1) = 0$.

Problem 9 (Sp95). Let $f_n: [0, 1] \rightarrow [0, \infty)$ be a continuous function, for $n = 1, 2, \dots$. Suppose that one has

$$f_1(x) \geq f_2(x) \geq f_3(x) \geq \dots \quad \text{for all } x \in [0, 1] \quad (*)$$

Let $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ and $M = \sup_{0 \leq x \leq 1} f(x)$.

1. Prove that there exists $t \in [0, 1]$ with $f(t) = M$.
2. Show by example that the conclusion of Part 1 need not hold if instead of $(*)$ we merely know that for each $x \in [0, 1]$ there exists n_x such that for all $n \geq n_x$ one has $f_n(x) \geq f_{n+1}(x)$.

Problem 10 (Sp95). Let n be a positive integer, and let $S \subset \mathbb{R}^n$ a finite subset with $0 \in S$. Suppose that $\varphi: S \rightarrow S$ is a map satisfying

$$\begin{aligned} \varphi(0) &= 0, \\ d(\varphi(s), \varphi(t)) &= d(s, t) \quad \text{for all } s, t \in S \end{aligned}$$

where $d(x, y) = \|x - y\|$ denotes the Euclidean metric. Prove that there is a linear map $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ whose restriction to S is φ .

Problem 11 (Sp95). Let n be a positive integer. Compute $\int_0^{2\pi} \frac{1 - \cos n\theta}{1 - \cos \theta} d\theta$.

Problem 12 (Fa81, Sp95). Let G be a subgroup of S_n of order p^k , where p is a prime not dividing n . Show that G has a *fixed point*; that is, one of the objects is left fixed by every element of G .

Problem 13 (Sp95, Sp19). Let y be a solution of $y''' - y = 0$ such that $y(t) \rightarrow 0$ as $t \rightarrow \infty$. Show that $y''(0) + y'(0) + y(0) = 0$.

Problem 14 (Sp95). Let $\mathbb{F}$ be a finite field, and suppose that the subfield of $\mathbb{F}$ generated by $\{x^3 \mid x \in \mathbb{F}\}$ is different from $\mathbb{F}$. Show that $\mathbb{F}$ has cardinality 4.

Problem 15 (Su82, Sp95). Let K be a nonempty compact set in a metric space with distance function d . Suppose that $\varphi: K \rightarrow K$ satisfies

$$d(\varphi(x), \varphi(y)) < d(x, y)$$

for all $x \neq y$ in K . Show there exists precisely one point $x \in K$ such that $x = \varphi(x)$.

Problem 16 (Sp95, Fa24). Let V be a finite dimensional vector space over a field $\mathbb{F}$, and let $L : V \rightarrow V$ be a linear transformation. Suppose that the characteristic polynomial χ of L is written as $\chi = \chi_1 \chi_2$, where χ_1 and χ_2 are two relatively prime polynomials with coefficients in $\mathbb{F}$. Show that V can be written as the direct sum of two subspaces V_1 and V_2 with the property that $\chi_i(L)V_i = 0$ for $i = 1, 2$.

Problem 17 (Sp95, Sp05, Sp15). Prove that there is no one-to-one conformal map of the punctured disc $G = \{z \in \mathbb{C} \mid 0 < |z| < 1\}$ onto the annulus $A = \{z \in \mathbb{C} \mid r < |z| < R\}$.

Problem 18 (Sp95). Let $\mathbb{F}$ be the finite field of cardinality p^n , with p prime and $n > 0$, and let $\text{GL}_2(\mathbb{F})$ be the group of invertible 2×2 matrices with coefficients in $\mathbb{F}$. Prove that $\text{GL}_2(\mathbb{F})$ has order $(p^{2n} - 1)(p^{2n} - p^n)$.

Problem 19 (Sp95). Let $\mathbb{F}$ be the finite field of cardinality p^n , with p prime and $n > 0$, and let G be the group of invertible 2×2 matrices with coefficients in $\mathbb{F}$. Show that any p -Sylow subgroup of G is isomorphic to the additive group of $\mathbb{F}$.